Quantification: 543_z13
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	9.5455	11	2.6216	count
ring_length_r30_px	headline	419.182	418	70.6199	px
ring_length_r60_px	reported	1073.27	1031	140.649	px
owned_length_px	ownership-dependent	3631.2	3390.74	1235.15	px
edge_count	ownership-dependent	167.182	136	56.5205	count
mean_edge_length_px	ownership-dependent	21.7765	21.2921	1.9078	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0500786	0.0487894	0.00716792	px^-1
ring_density_r60	normalised	0.0491689	0.0492474	0.00537553	px^-1
roots_per_100px_perimeter	normalised	4.6785	4.2451	1.1799	per 100 px
ring_attached_length_r30_px	network v2	392.273	374	75.3924	px
ring_attached_length_r60_px	network v2	942.273	898	190.567	px
ring_length_w_r30_px	network v2	476.621	466.664	80.192	px
ring_length_w_r60_px	network v2	1218.82	1149.67	161.988	px
sholl_crossings_r10	network v2	10	10	2.4083	count
sholl_crossings_r20	network v2	13.8182	14	2.9939	count
sholl_crossings_r30	network v2	13.9091	14	2.468	count
loop_count	loops, report only	17.8182	15	9.8875	count
loop_edge_share	loops, report only	0.3873	0.424	0.1027	share
ring_width_mean_r30_px	width	7.2019	7.3203	0.49	px

Per field
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measure	value	units
analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	44313	px
canalicular_length_density_per_px	0.0431021	px^-1
junction_count	1096	count
junction_density_per_px2	0.00106605	px^-2
lacuna_count	11	count

lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	192	count
mean_component_length_px	230.797	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0490276	px^-1
field_density_without_flagged_per_px	0.0431021	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	256	px
unattached_component_count	13	count
canalicular_length_density_attached_per_px	0.0431021	px^-1
field_length_density_with_unattached_per_px	0.0433511	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	245	count
loop_edge_share	0.4275	share
edge_count	2241	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.803	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	44313	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	357	975	4395.66	215	20.4449	7.4298
2	11	422	1031	5471.08	238	22.9877	6.9389
3	11	401	960	2481.41	118	21.0289	6.8641
4	5	305	953	2533.76	119	21.2921	7.6428
5	11	418	1155	5107.46	233	21.9204	7.6884
6	6	415	1014	2205.93	121	18.2308	7.3203

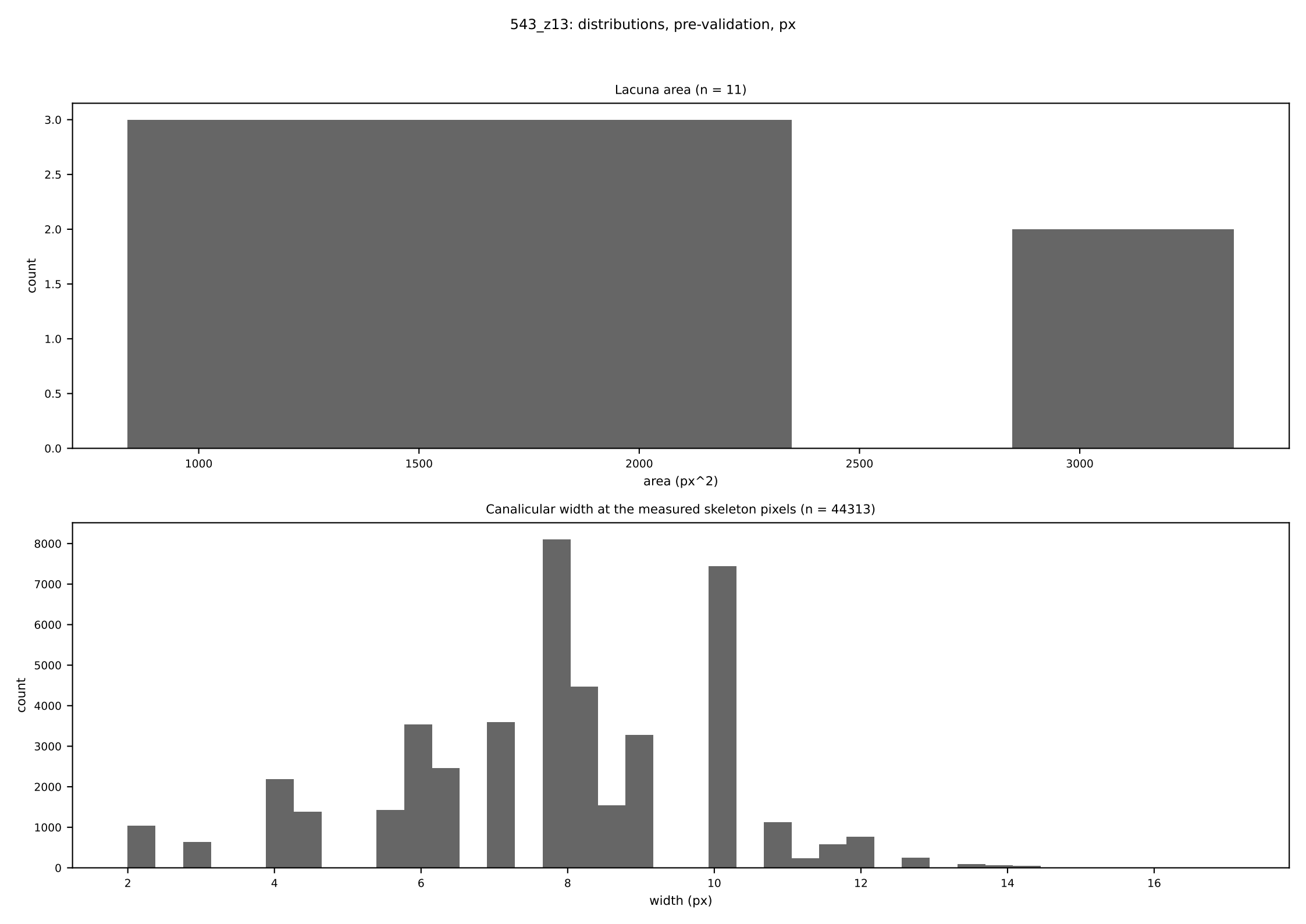
7	13	504	1238	4665.67	220	21.2076	6.4821
8	10	519	1198	3390.74	132	25.6874	7.1878
9	7	332	867	2082.83	90	23.1426	7.3689
10	11	431	1090	3110.17	136	22.8689	6.3846
11	12	507	1325	4498.55	217	20.7306	7.9136

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	205.037	8499	22530	1	1	0.0420049	0.0432756	3.9017
2	151.297	6982	17707	1	1	0.0604411	0.0582256	7.2705
3	205.037	8219	20045	1	1	0.0487894	0.0478922	5.3649
4	139.681	6734	18999	1	1	0.0452926	0.0501605	3.5796
5	223.823	8900	23453	1	1	0.0469663	0.0492474	4.9146
6	141.338	6516	17817	1	1	0.0636894	0.0569119	4.2451
7	254.936	9810	24446	1	1	0.0513762	0.0506422	5.0993
8	278.593	10387	26403	1	1	0.0499663	0.0453736	3.5895
9	195.723	8083	21794	1	1	0.0410739	0.0397816	3.5765
10	184.995	7835	21305	1	1	0.0550096	0.0511617	5.9461
11	301.806	10961	27498	1	1	0.0462549	0.0481853	3.9761

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	355	909	422.274	1129.74	11	12	10
2	397	898	466.664	1149.67	9	14	15
3	334	640	456.818	1090.28	12	14	13
4	300	858	351.078	1078.61	6	9	13
5	374	1022	467.706	1290.65	10	14	17
6	370	824	464.392	1138.12	7	13	14
7	504	1161	577.316	1410.9	13	17	14
8	505	1173	593.416	1377.38	13	17	18
9	297	774	376.664	991.365	9	9	11
10	395	844	484.848	1235.18	8	15	12
11	484	1262	581.659	1515.11	12	18	16


543_z13: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 44313)
count
width (px)